

BUDDHSEN TRIPATHI

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EDUCATION

New York University

Master of Science in Computer Science

Sep 2025 – May 2027

New York, USA

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering

Jul 2019 – May 2023

Vellore, India

EXPERIENCE

New York University

Course Assistant, Software Engineering / Research Collaborator

Sep 2026 – Present

New York, NY

- Assist students with software engineering concepts, assignments, and debugging; evaluate coursework and provide feedback on software design, testing, and engineering best practices.
- Collaborate with faculty on LLM coding agent reliability research, developing an HMM-based framework to track semantic drift and conducting test-grounded evaluations of long-horizon code repair and regression patterns.

Amadeus

Software Development Engineer I

Jul 2023 – Aug 2025

Bengaluru, India

- Optimized document generation systems in Java Spring, Freemarker, and Angular, reducing rendering latency by 12% across thousands of daily airline transactions.
- Migrated 75% of on-premises services to Azure with CI/CD pipelines and containerization, achieving 99.9% uptime and improved fault tolerance.
- Resolved 200+ production incidents and problem reports, remediating critical security findings through root-cause analysis and refactoring to reduce vulnerabilities by 50%.

PROJECTS

DeepFind.Me (Acquired by UserSearch.com)

Next.js, NestJS, PostgreSQL, Docker, AWS, OpenAI API

- Built and scaled an AI-powered Open-Source Intelligence (OSINT) platform integrating 30+ intelligence tools, reaching 5K+ users and 30K+ monthly visits.
- Engineered an autonomous research agent orchestrating 15+ OSINT tools through multi-step planning, dynamic tool selection, and sub-agent delegation to execute complex investigations.

Scaling Laws for Music Language Models

Python, PyTorch, NumPy, SciPy, Matplotlib

- Built an end-to-end ML pipeline converting 120K+ MIDI files into ABC notation with preprocessing, cleaning, and tokenization.
- Trained and benchmarked Transformer and LSTM language models (1M–100M parameters) under identical compute budgets to study scaling laws in symbolic music generation.

HACKATHONS

1st Place - vibefORWARD: M2 - Agents Hackathon (Jun 2026): *FRTC* - autonomous fraud-ring investigator coordinating six specialist agents through a shared memory graph to analyze 90 days of transactions in real time.

2nd Place - Agentic Engineering Hack @ Datadog HQ (May 2026): *Denial Rescue* - 7-agent healthcare-regulatory legal system that issue-spots payer denials and HIPAA breaches against a 1,900-authority federal health law corpus in 90 seconds.

1st Place - vibefORWARD Hackathon (Apr 2026): *AI Blox* - visual drag-and-drop AI engineering tool with 500+ blocks for LoRA fine-tuning, RAG pipelines, and multi-agent orchestration, generating production-ready code.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, Go, C++

Frameworks: FastAPI, Next.js, React, Node.js, NestJS, Spring Boot, Angular

Databases: PostgreSQL, MySQL, MongoDB

AI & ML: PyTorch, TensorFlow, scikit-learn, LangChain, RAG, fine-tuning, embeddings

DevOps & Tools: Docker, Git, GitHub Actions, Jenkins, JUnit